DATA EXTRACTION · VERIFICATION

Experimental results of the ECG scoring system

Only source: the attached PDF (Cardiol Res. 2016;7(3):110-116)

3	9	50	0
tables used	ROC thresholds (Table 7)	validation (Table 8)	transcription errors

Scope

*Experimental results are in **Tables 6, 7 and 8** and in the summary metrics (Abstract, Results, Conclusion). Tables 1 to 5 describe the patients, echo and ECG findings, and how the score was built, so I left them out.*

1 Table 7: ROC cut-off points (p. 114)

Method	Dataset	Evaluation metric	Result	Page	Remarks
ECG scoring system (LAH, wide QRS, RBBB, ST-T changes, prolonged QT)	not stated for the ROC analysis (HF patients, n = 110 elsewhere)	Table 7, threshold ≥ -2.0000: Sensitivity / 1 - specificity / Specificity	**1.000 / 1.000 / 0.000**	114	Dataset not named for Table 7
same	same	Threshold ≥ -0.5000	**1.000 / 0.980 / 0.020**	114	
same	same	Threshold ≥ 0.5000	**1.000 / 0.700 / 0.300**	114	
same	same	Threshold ≥ 1.5000	**0.950 / 0.520 / 0.480**	114	
same	same	Threshold ≥ 2.5000	**0.867 / 0.260 / 0.740**	114	
same	same	Threshold ≥ 3.5000	**0.700 / 0.040 / 0.960**	114	Corresponds to the paper's cut-off of 4 (my reading)
same	same	Threshold ≥ 4.5000	**0.483 / 0.020 / 0.980**	114	
same	same	Threshold ≥ 5.5000	**0.117 / 0.000 / 1.000**	114	
same	same	Threshold ≥ 7.0000	**0.000 / 0.000 / 1.000**	114	

2 Validation at cut-off 4 (p. 110, 113, 115)

Method	Dataset	Evaluation metric	Result	Page	Remarks
ECG scoring system, cut-off 4 (+4 to +6 = HFrEF, -1 to +3 = HFpEF)	Validation sample, n = 50	Sensitivity	**76%**	110, 113, 115	Differs from 0.700 in Table 7 (see below)
same	same	Specificity	**96%**	110, 113, 115	
same	same	Positive predictive value	**95%**	110, 113, 115	
same	same	Negative predictive value	**80%**	110, 113, 115	
same	same	Accuracy	**86%**	110, 113, 115	
same	same	Table 8 counts, score +4 to +6	**19** (EF ≤ 40%), **1** (EF > 40%)	115	Printed as “+4 SD +6”
same	same	Table 8 counts, score -1 to +3	**6** (EF ≤ 40%), **24** (EF > 40%)	115	Printed as “-1 SD +3”

3 Table 6: probability of HFrEF (p. 113)

Method	Dataset	Evaluation metric	Result	Page	Remarks
ECG scoring system, probability model	not stated	Probability of HFrEF (%), score -1	**0.9**	113	Table 6
same	same	Probability of HFrEF (%), score 0	**3.16**	113	
same	same	Probability of HFrEF (%), score 1	**10.4**	113	
same	same	Probability of HFrEF (%), score 2	**29.3**	113	
same	same	Probability of HFrEF (%), score 3	**59.6**	113	
same	same	Probability of HFrEF (%), score 4	**84**	113	
same	same	Probability of HFrEF (%), score 5	**94.9**	113	
same	same	Probability of HFrEF (%), score 6	**98.5**	113	

4 Sources used

Table 6 (p. 113) **Table 7 and Figure 1** (p. 114) **Table 8** (p. 115)

Summary metrics in the Abstract (p. 110), Results text (p. 113) and Conclusion (p. 115)

5 Verification against the original

Values: every number matches the corresponding table cell in the document text, with the same decimals and units. No transcription discrepancy found.

Check	Calculation	Verdict
Table 7: “1 - specificity” + “Specificity”	= 1 in all 9 rows	**OK**
Table 8: sensitivity	19/25 = 76%	**OK**
Table 8: specificity	24/25 = 96%	**OK**
Table 8: positive predictive value	19/20 = 95%	**OK**
Table 8: negative predictive value	24/30 = 80%	**OK**
Table 8: accuracy	43/50 = 86%	**OK**
Table 7: fit with group sizes	0.700 = 42/60; 0.483 = 29/60; 0.960 = 48/50	Suggests n = 110 (never stated)

6 Ambiguities and inconsistencies

Sensitivity mismatch	At the cut-off of 4, Table 7 gives sensitivity 0.700, while the validation gives 76%. Specificity is 0.960 in Table 7 and 96% in the validation. The paper does not explain the difference.
Unclear validation sample	The text says “validated in several samples”, but Table 8 has n = 50. It is not stated whether these 50 patients are separate from the 110 or a subset. The main cohort also has 50 HFpEF patients, which could cause confusion.
“SD” in Table 8	“+4 SD +6” and “-1 SD +3” are most likely “+4 to +6” and “-1 to +3” (the text says so), but the printed “SD” is unexplained.
Table 7 columns	“Sensitivity” appears twice, and the second Sensitivity/Specificity pair repeats the plotted data. Thresholds are half-points (for example 3.5), while the text speaks of a cut-off of 4.
Figure 1	The x-axis shows row numbers 1 to 9, not score values. The legend has a typo (“spesificity”), and values are only approximate when read off the curve.
No Dataset or Method names	This is a clinical study, not a machine-learning benchmark, so “Method” and “Dataset” fit loosely. I used the scoring system and the patient sample.

Next step
Checked against the text extracted from the PDF. A final visual comparison of Tables 6 to 8 in the original PDF is still worthwhile.